

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

Once the DAC operation is confirmed, the remaining circuit can be assembled and tested. Before connecting the Raspberry Pi, ensure that there are no short circuits on the board. Do not plug in the shift-register IC U1 yet. Connect a +5V power supply from a standard laboratory power supply and check that the LED D1 glows properly. Measure the voltage at IC pins 16 (+5V) and pin 8 (GND/0V). Once everything is satisfactory, plug in the 74HC595 shift-register IC, and initiate the connections to the Raspberry Pi board.

Ensure that the Pi board is powered off, then connect the power

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supply pins +3.3V, +5V, and GND from the Pi board to the waveform generator PCB. Check for any short circuits. Then connect the remaining GPIO pins. Verify all connections and power up the Pi board.

Using the Python programming environment on the Pi, load the Python program for triangular wave generation and run it. The output waveform at connector J1 – ‘OUT’ can be observed on an oscilloscope.

Once this signal is observed correctly, test programs for other waveforms, such as sine, square, forward ramp, and reverse ramp. Users can also write their own programs to generate arbitrary waveforms for DC voltages, as needed.

During testing, it was observed that the update speed from the Raspberry Pi is somewhat slow, resulting in a very low frequency or rate of these waveforms in the

sub-Hz region. This limitation might be due to the existing libraries and functions supported within the Python environment on the Pi board.

Fig. 7 (A to F) illustrates the various waveforms generated using this DAC board and shows how the shift-register IC receives data from the SPI port of the Pi board. **EFY**

The author, Prasanna Waichal, holds a Master's Degree in Electronics and has nearly 30 years of work experience in industrial R&D for the design and development of specialised and strategic electronics systems, as well as teaching and research at postgraduate levels both in India and abroad. His passions include radio receiving and studying solar activities and their effects on the ionosphere. He is also a licensed Amateur Radio Operator with the call sign VU3QOI.

Co-authors Arpita Waichal, Nikita Thorat, and Pragati Todkar are currently pursuing their B.Tech in Electronics and Telecommunication Engineering at DKTE's College of Engineering, Rajwade, Ichalkaranji, Kolhapur, Maharashtra



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